



RefillRadar

Group 2

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Introduction

What is RefillRadar?

Managing medications shouldn't feel like a full-time job. But when you're juggling different dosages, schedules, and refill requirements, it quickly becomes overwhelming.

That's where RefillRadar comes in: a simple, intuitive app designed to help users stay on track with prescriptions.



Competitor Analysis

 = Strong
 = Okay or limited
 = Weak or missing

FEATURES	RefillRadar	Medisafe	Dosecast	Mango Health
Cost / Subscription Model	Free 	Pay for premium features 	Paywall 	Free 
Ease of Use	Clean, minimal 	Intuitive but busy 	Compressed UI 	Friendly design 
Navigation Clarity	Clear layout 	Tabs & menus 	Dense options 	Multilayered 
Customization Options	Color + icon picker 	Limited personalization 	Only a few settings 	Points system only 
Refill Reminders / Notifications	WIP  	Works well 	Limited free alerts 	Fully Active 

Designed for Humans. Built with HCI Principles.

✓ **Consistency**

Uniform buttons, spacing, icons across all screens

✓ **Visibility of System Status**

Instant feedback: highlights, borders, and dialogs

✓ **Error Prevention & Recovery**

Smart filters, dropdowns, and confirmation prompts

✓ **Flexibility & Efficiency**

Scrollable layouts, fast access on any screen size

✓ **Affordance & Discoverability**

Labeled icons, hint text, pill-button selectors

✓ **Accessibility & Simplicity**

Big touch targets, clean fonts, minimal flows

The screenshot shows a mobile application interface for a 'Prescription' form. The title bar at the top is dark grey with three colored window control buttons (red, yellow, green) on the left and the word 'Prescription' in white. Below the title bar is a white header with a blue back arrow and the text 'Edit Prescription'. The form is divided into several sections: 'Category' with a row of six circular icons (the first icon, containing a pill, is highlighted with a blue square); 'Color' with a row of seven colored circles (the last circle, light blue, is highlighted with a blue square); 'Medication Name' with a text input field containing 'adderal 30'; 'Dosage' with a dropdown menu (the dropdown is open, showing options 'HOUR(S)', 'DAY(S)', and 'WEEK(S)', and is highlighted with a blue rectangle); and 'Start Date' with a calendar icon. Blue arrows originate from the HCI principles on the left and point to these specific UI elements: from 'Consistency' to the pill icon; from 'Visibility of System Status' to the pill icon and the color selection; from 'Error Prevention & Recovery' to the dosage dropdown; from 'Flexibility & Efficiency' to the dosage dropdown; from 'Affordance & Discoverability' to the dosage dropdown; and from 'Accessibility & Simplicity' to the 'Start Date' field.

App Traversal

01

Create Account

Simple sign-up and sign-in.

Just name, email, and password for a quick setup.

02

Home Screen

Displays all user medications for easy access.

User-friendly and intuitive layout helps guide you through multiple features.



03

New Medication

Tap the “+” button to start adding a new prescription.

Users can assign colors and icons to help identify meds and indicate their reminder preferences.



04

Inspect/Edit Medication

Tap on a medication card to see and edit its information.

Users can adjust any detail, especially useful for continual, long term medications.



05

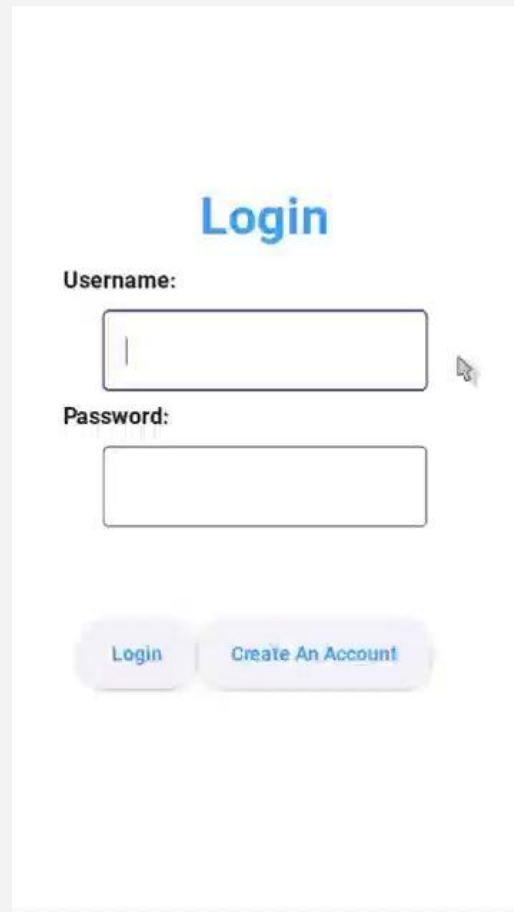
Calendar View

Helps visualize upcoming refills and reminder dates.

Users can see on a calendar what medications are about to run out and when their reminders are set to notify them in advance.

Sign In & Sign Up

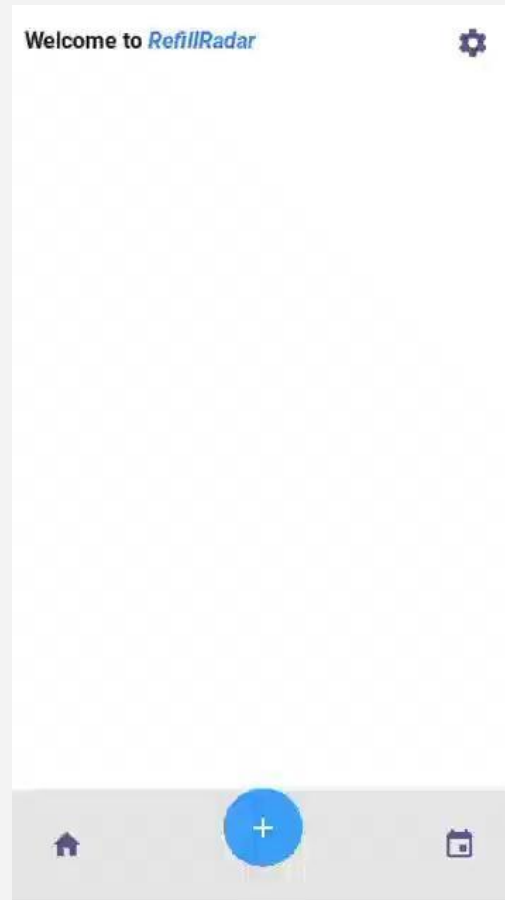
- We designed the login and sign-up process to be quick and simple, with just a few steps—perfect for users who may not be tech-savvy.
- We plan to add email-based features like two-factor authentication and account recovery to improve security without sacrificing convenience.



The image shows a clean, minimalist login interface. At the top, the word "Login" is displayed in a bold, blue font. Below it, the label "Username:" is followed by a rectangular input field. A mouse cursor is positioned over the right side of this field. Underneath the username field, the label "Password:" is followed by another rectangular input field. At the bottom of the form, there are two rounded rectangular buttons. The first button is light blue with the text "Login" in a darker blue. The second button is light gray with the text "Create An Account" in a darker gray.

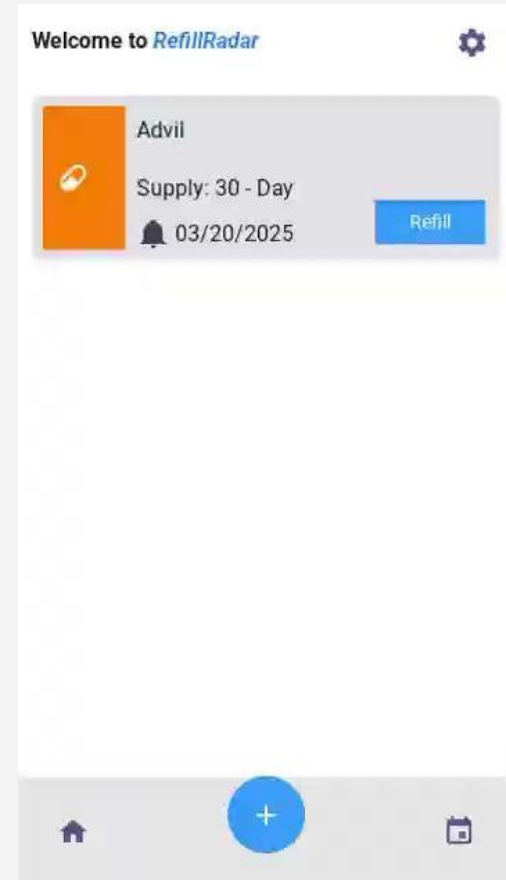
Add New Prescription

- Tapping the plus button on the home screen lets users quickly add a new prescription.
- They just enter the medication details and their refill preferences, then hit 'Add'—and everything gets saved to the database instantly.”



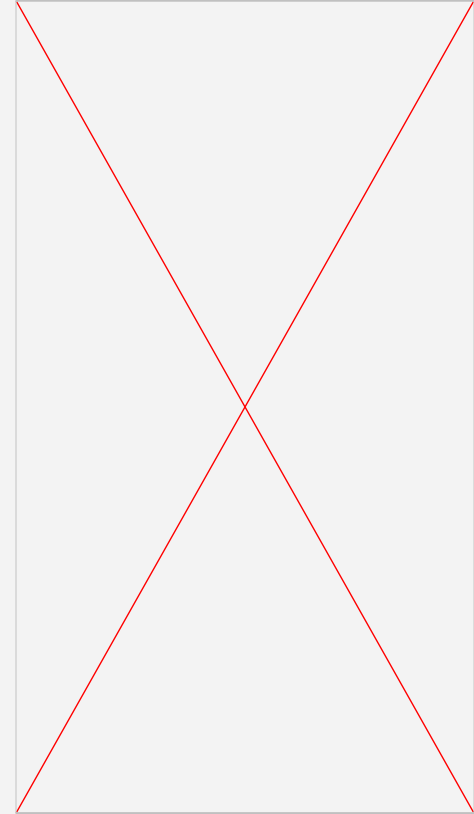
Home Screen & Med Info

- Users can see key info like supply and first reminder date at a glance.
- Tapping “Refill” changes the button to “Refilled” for quick tracking.
- In future versions, this will reset the med’s start date (prescription rollover).
- Icons and color coding help users easily recognize meds and reduce cognitive load.



Calendar View & Settings

- The calendar icon takes users to a screen that highlights upcoming refill dates and supply end dates.
- While the backend isn't fully connected yet, this demo shows how reminders will appear as blue highlights.
- Tapping a date brings up a popup with scheduled reminders or supply info.
- Users can also access the **Logout** option here.
- We plan to add a **Delete User Data** button in the future for privacy and security (like switching devices or account issues).



User Testing & Insights

To test the usability and effectiveness of our app, we conducted user surveys with friends and family and observed them traverse the app on their own accord.

User Consensus

- The app's purpose was clear
- Expected features were present
- Pathways were intuitive
- Simple and accessible
- Actions didn't take too many layers of menus to achieve

Tester Diversity

- Age (22 - 70)
- Varying levels of tech savvy
- Take 1+ medications

Common Critiques

- Lack of clarity for mandatory fields
- Unclear what fields could autofill
- Users requested a visual indicator for the current day on the calendar

Challenges and Looking to the Future

Challenges

- First time doing mobile app development
- Issues with packaging app code to be ran on an emulator
- Kivy was difficult to work with for frontend development

For the Future

- Test on Emulator - For mobile notifications
- Connect Calendar - Backend + highlight today
- Enable Refills - Hook up rollover refill button logic
- Finish Settings - Connect delete data option to backend
- Add Features - Export data, pharmacy tools, dosage reminders

Tech Stack

- **Frontend / UI:**

- Kivy / KivyMD – For build a responsive mobile-ready UI.

- **Programming Language:**

- Python 3.13 – Core language for logic, UI, and backend functions.

- **Database & ORM:**

- SQLite – Local prescription data storage.
- SQLAlchemy – Object Relational Mapper for managing data with Python objects.

- **Security & Encryption:**

- cryptography.fernet – Encrypts sensitive data like passwords.
- hashlib / base64 / secrets – For hashing, encoding, and token generation.

- **Environment Configuration:**

- dotenv (.env) – Stores sensitive variables (e.g., keys) outside the main code.

- **Dependency Management:**

- requirements.txt – Tracks all needed libraries for setup.

- **Version Control & Collaboration:**

- Git & BitBucket – For versioning, managing branches, and team collab.
- GitKraken – Visual Git tool for merging, and reviewing pull requests.

- **IDE:**

- Visual Studio Code – Used for writing, testing, and debugging code.



QUESTIONS